

Research Topic and Methodology Justification

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1. Research Topic

Topic: Using Vision-Language Models to support spatiotemporal reasoning for safety monitoring and activity tracking on active construction sites.

Construction sites rank among the most hazardous occupational environments globally. They are characterized by continuous, often unpredictable interactions among workers, heavy equipment, and materials across a spatially and temporally evolving environment. Incidents arising from undetected unsafe behaviors, such as personal protective equipment (PPE) non-compliance, unauthorized proximity to machinery, or deviations in task sequences, carry severe human and economic consequences. At the same time, accurate progress tracking remains a persistent operational challenge: manual inspection and reporting introduce delays and inconsistencies that compound project delivery risk. This review examines how Vision-Language Models (VLMs) and related multimodal architectures have been applied to these two intertwined problems within active construction environments and their simulated or laboratory-reconstructed equivalents. The topic sits at the intersection of construction management, computer vision, and natural language processing, making it relevant to researchers and practitioners across all three communities.

2. Research Questions and Purpose

The purpose of this review is to systematically map the current state of Vision-Language Model research as applied to spatiotemporal reasoning in construction—identifying prevalent architectures, data fusion strategies, deployment barriers, and evaluation frameworks, and locating the gaps where future primary research is most needed. To operationalize this purpose, the review is structured around four specific research questions (RQs):

RQ1: What are the prevalent VLM architectures currently utilized for spatiotemporal analysis in construction environments?

RQ2: How is multimodal data—visual and textual—fused and processed to evaluate safety hazards on active construction sites?

RQ3: What are the primary methodological bottlenecks in deploying these models for real-time progress tracking and safety monitoring?

RQ4: How is the performance of VLMs currently evaluated and benchmarked for spatiotemporal tasks in construction contexts?

Together, these four questions guide a structured mapping of methods, datasets, evaluation practices, and deployment realities across the existing literature.

3. Selected Review Methodology

Methodology: Scoping review, conducted in accordance with PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews) guidelines.

A scoping review is a form of knowledge synthesis that systematically locates and charts the available evidence across a defined topic area. Rather than aggregating effect sizes or appraising the methodological quality of individual studies—as a systematic review or meta-analysis would do—a scoping review prioritizes breadth, comprehensive coverage, and transparent, reproducible reporting. The goal is to characterize the range and nature of the existing evidence, identify where the literature is sparse or inconsistent, and clarify how key concepts and methods are being used across disciplines. The PRISMA-ScR framework provides a structured checklist and flow diagram that makes every stage of the search, screening, and selection process fully auditable.

4. Methodology Justification

The scoping review methodology is the most defensible and appropriate choice for this topic and research questions for two principal reasons, in addition to several practical considerations.

Alignment with mapping-oriented research questions. All four research questions ask “what” and “how”—they seek a structured map of architectures, fusion strategies, bottlenecks, and evaluation frameworks rather than a quantitative estimate of a particular intervention effect. Scoping reviews are purpose-built for this kind of inquiry. A traditional meta-analysis would be premature at this stage because the primary objective here is to chart the extent, range, and nature

of an emerging architectural family, not to synthesize narrow intervention effects across comparable controlled experiments.

Accommodation of methodological heterogeneity. Evaluation metrics across the relevant literature vary substantially: from geometric precision metrics such as mean Average Precision (mAP), used in object-detection studies, to natural language generation metrics such as BLEU and CIDEr, used in image captioning and visual question-answering studies. This heterogeneity renders a standardized statistical meta-analysis inappropriate. A scoping review explicitly accommodates diverse study designs and reporting conventions within one coherent synthesis framework, without requiring methodological equivalence as a precondition for inclusion.

Breadth of literature included. Because the topic spans construction management, computer vision, and natural language processing, relevant studies include empirical field deployments, simulation-based benchmarks, dataset introduction papers, and foundational architectural comparisons. The scoping methodology permits inclusion of all these study types, which is essential for capturing the full landscape of a multidisciplinary and fast-moving field.

Practical considerations. The PRISMA-ScR framework makes the review transparent and reproducible within the constraints of this course. The structured search strings, two-stage screening procedure, and flow diagram documentation can be executed using the databases available through the university library, within the available timeframe, and with results that are fully auditable by an external reader.

5. Reflection on the Search Process

Three significant developments have shaped this project since the initial proposal. First, in response to instructor feedback from Week 1, the original broad framing—VLMs in built environments generally—was narrowed to focus specifically on safety monitoring and activity tracking during the active construction phase. This narrowing made the research questions more precise and the inclusion criteria more coherent and defensible. Second, a single broad research question was replaced with four operationally distinct questions (RQ1–RQ4) that distribute the mapping task across architectures, fusion strategies, deployment barriers, and evaluation frameworks. This structure makes the eventual synthesis more organized and the contribution of the review clearer.

to a reader. Third, the adoption of the PRISMA-ScR reporting framework emerged as a natural commitment once the scoping methodology was confirmed, because it provides the reproducibility scaffolding that any peer-reviewable evidence synthesis requires and formalizes what would otherwise be informal methodological choices.